FOMO

Follow-up Observations of Moving Objects

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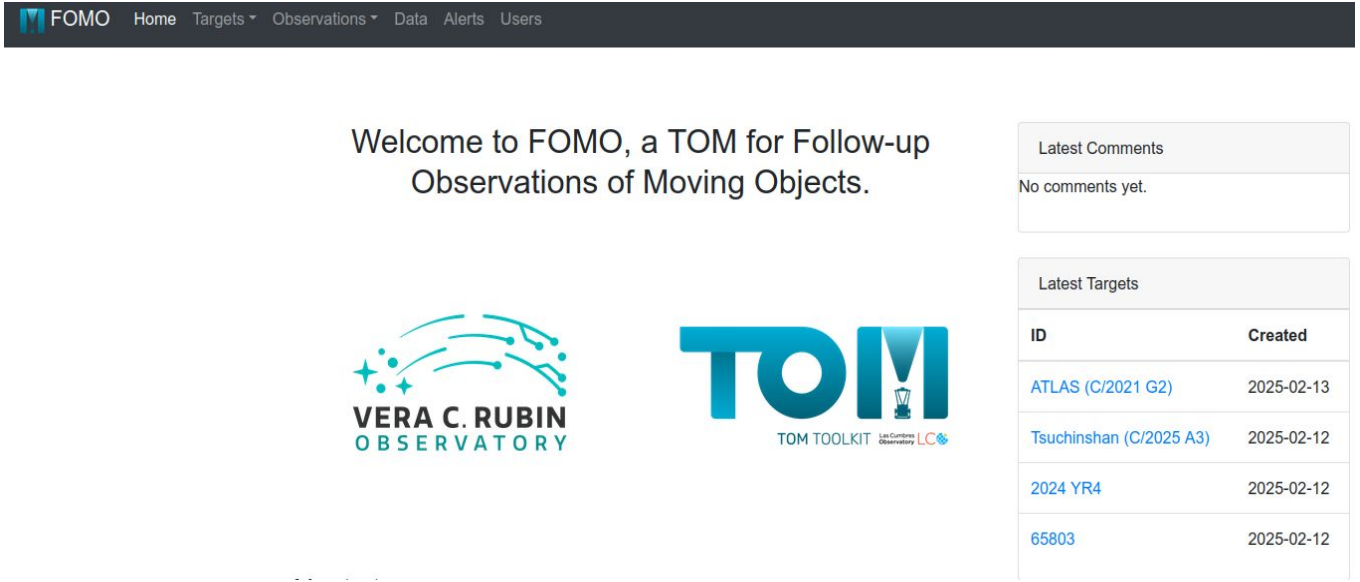
What is FOMO?

- A centralized system to **coordinate follow-up observations** between community members
- A single place for **overview of proposals and facilities** for ToO observations of exceptional objects
- **Tools to plan and optimize observations** across facilities
- **Automation** to handle the increased target load efficiently
- A single reference point for **follow-up status and observation history**
- **Better tracking** from alert → observation → data → publication

TOM-Toolkit

Our base; a software package/template built by LCO to facilitate observing projects and collaborations for astronomers without high level database/software expertise



Used for:

- Supernovae
- Microlensing
- ZTF targets
- And more!

TOM-Toolkit

What they have so far:

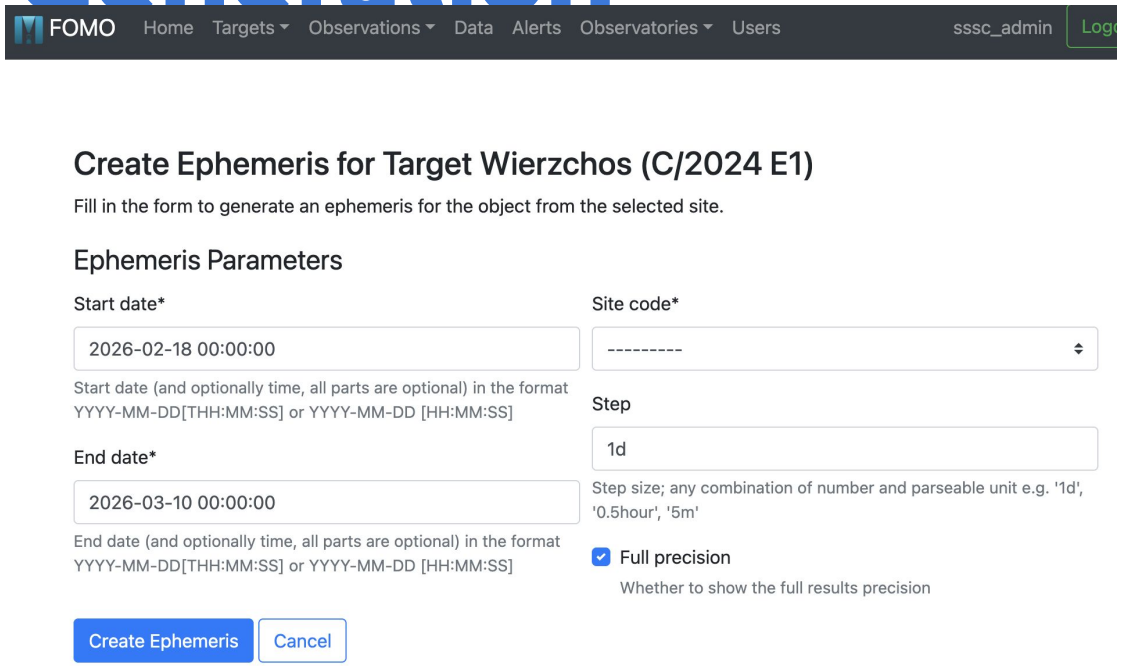
- Ingest of target info from brokers, MPC and now JPL APIs
- Observation scheduling on LCO, Gemini, SOAR
- Observation status updates and data ingest

What they are working on:

- RSP support
- ESO support
- Facility status
- Enhanced user registration

The team is very collaborative and working with us to achieve shared goals for the toolkit/FOMO

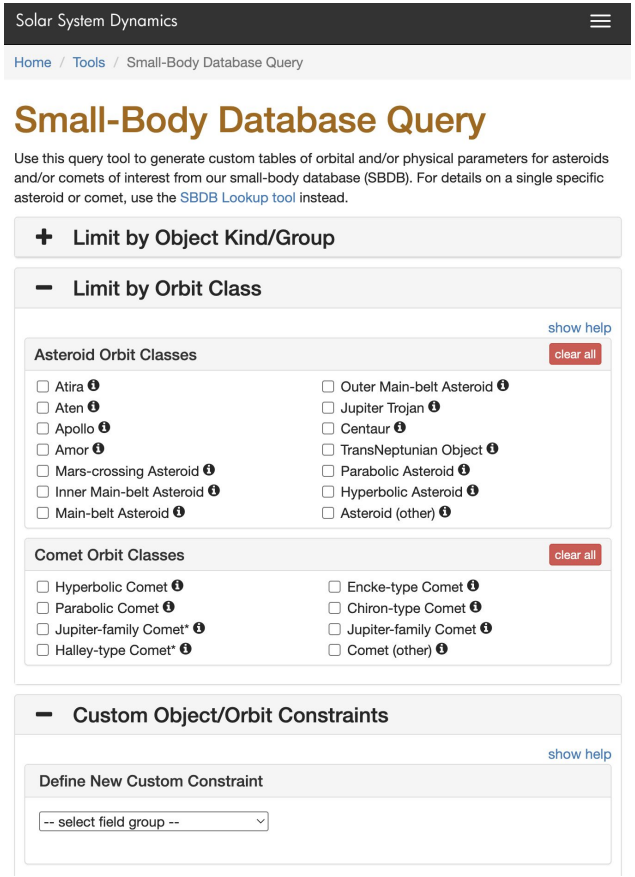
Updates: Ephemeris Generation



- Uses custom code to turn TOM `Targets` into input for Sorcha
- Uses pieces pulled from Sorcha plus additional code to generate observational details

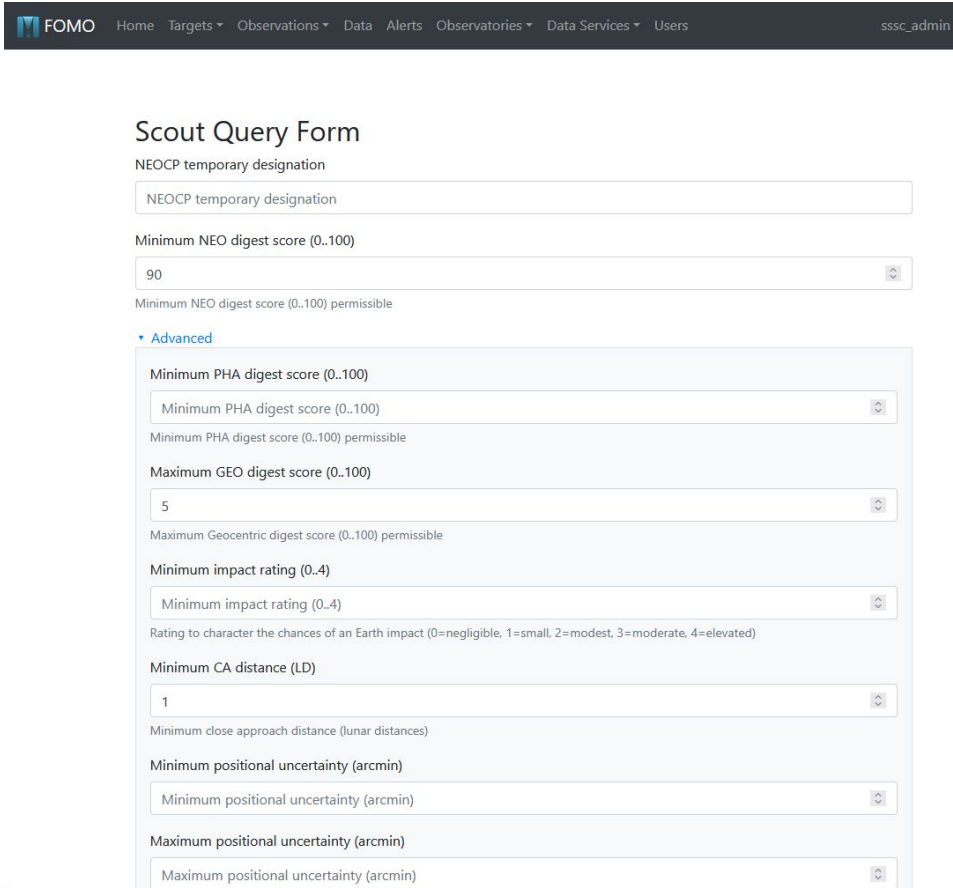
Updates: JPL SBDB Ingestion

- Ingestion of new, exciting discoveries - after MPC designation
- Ingest groups of targets according to cuts/filters on the JPL parameters
- Will fetch new objects daily through periodic (cron) tasks



Updates: Scout Ingestion

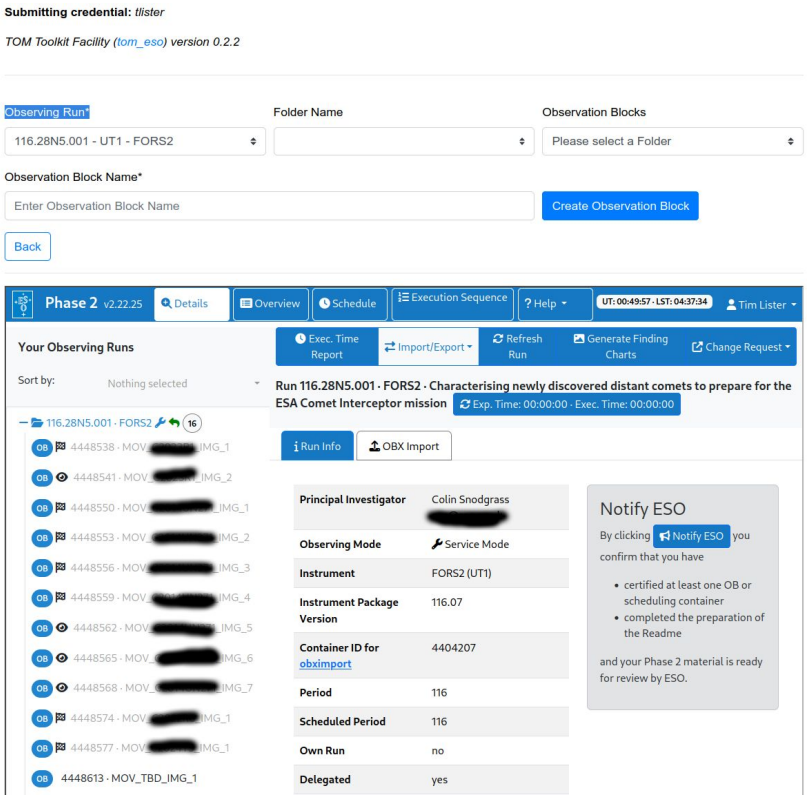
- (Optionally) filterable form and backend to ingest NEOCP objects from Scout
- Working now but needs some additional work to fetch, store and version metadata e.g. *neoScore*, impact rating etc
- Code review and merge into TOM Toolkit to happen RSN...



FOMO: Follow-Up Observations of Moving Objects

Updates: ESO Integration

- Connects FOMO to ESO Phase 2 interface
- Can fetch and create basic Observation Blocks (OBs)
- Can easily add auto generated PAF ephemeris file
- Would love feedback from ESO observers on what else is needed (favourite instruments, modes, template filling etc)



Current Status

Adding targets: manually, JPL query

Retrieving target data: automatic from JPL/MPC

Sending robotic observation requests: LCO, Gemini, SOAR, ESO

Planning: ephemeris generation module

Status updates of requests

We have successfully manually ingested a target and sent an observation requests and received status updates and final data products.

Minimum Product

Adding targets: manually, JPL Query, **SCOUT query, broker alerts**

Retrieving target data: automatic from JPL/MPC, **broker(s)**

Sending robotic observation requests: LCO, Gemini, SOAR, ESO

Planning: ephemeris generation module

Status updates of requests

Management of users/proposals with different permissions

Front-end design + new logo?

Important Near-Future Additions

Adding targets: manually, JPL Query, SCOUT query, broker alerts

Retrieving target data: automatic from JPL/MPC, broker(s)

Sending robotic observation requests: LCO, Gemini, SOAR, ESO

Planning: ephemeris generation module, many more helpful tools!

Status updates of requests

Management of users/proposals with different permissions

Front-end design + new logo?

Integration with manual facilities (!!!!)

Feedback Requested!

What key components might we have missed?

Are there critical functionalities we haven't considered?

Your insights will help us improve the system - thank you!

GitHub https://github.com/lsst-sssc/fomo

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Welcoming Collaborators!

- Skymaps and Rubin schedule viewer
- Exposure Time Calculators
- Target and alert ingesters
- Front-end design
- Any idea(s) you may have?

Interested in helping out?

Please reach out!

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